Mayank Mishra

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.03
St. Xavier's Senior Secondary Co-ed School, BHEL, Bhopal	Class XII (CBSE)	2019	78.80%
St. Xavier's Senior Secondary Co-ed School, BHEL, Bhopal	Class X (CBSE)	2017	79.80%

PROJECTS

Dual-Stream Synthetic Scene Classifier | *IITH Deep Learning Hackathon PyTorch, OpenCV, NumPy, pytest*

- Built a dual-stream CNN trained entirely from scratch (no pre-trained weights) combining a raw-RGB stream with a custom material-invariant edge-feature stream, reaching 0.8092 public leaderboard accuracy -- up from a 0.7534 single stream baseline
- Designed an edge pipeline (bilateral filtering + Canny/Sobel/Laplacian) to strip specular-highlight shortcuts from metallic objects, paired with a MixUp- and SWA-regularized RGB stream on a shared ResBlock+SE backbone
- Closed the train/leaderboard gap via two-round pseudo-label self-training, BatchNorm test-time adaptation, and 4-way deterministic TTA, iterating through 5 ablations for a cumulative +5.6pp gain over the baseline
- Refactored the notebook pipeline into a modular, pip-installable Python package with a pytest suite, published on GitHub

Multilingual Business Entity Resolution | *Amazon ML Challenge 2026 Python, PyTorch, LightGBM, Transformers, scikit-learn*

- Engineered a multilingual retrieval pipeline that matches noisy business records across 3 sources with no shared identifiers. It combines character 3-gram TF-IDF, exact-key blocking and a fine-tuned multilingual-e5-small kNN search, with Indic-script transliteration
- Fine-tuned 3 cross-encoders (XLM-RoBERTa-large, mDeBERTa-v3, BGE-reranker-v2-m3) on held-out data. Their scores go into a two-stage LightGBM ranker with context features (per-record rank, competition between records, agreement between related candidates)
- Designed a decision layer that applies a one-to-one assignment and then picks the match set that maximises expected F0.5. A LightGBM pre-filter cuts the candidates to about 5 per record, which keeps inference fast and precise
- Generalized to France, a country absent from the training data, by fine-tuning on synthetic French pairs made from labelled US data. This beat self-training and proxy validation. The final score was 0.98964 F0.5, ranking 63rd among 32,000+ teams

FastAPI Docs RAG -- RAG System with an Automated Evaluation Harness | *Python, FastAPI, Qdrant, Google Gemini, DeepEval, Docker, GitHub Actions*

- Built a retrieval-augmented generation pipeline that answers questions over FastAPI's docs and source code, using content-aware chunking and Qdrant vector search
- Designed a rigorous automated evaluation harness (DeepEval: faithfulness, answer relevancy, contextual precision/recall) scored against a 437-example golden dataset built from real, accepted-answer GitHub Discussions
- Improved retrieval quality by implementing Anthropic's Contextual Retrieval technique and evaluating a cross-encoder reranking stage through systematic ablation studies
- Deployed as a live, publicly accessible Dockerized FastAPI service with a CI regression gate (GitHub Actions) guarding every retrieval/generation change, plus Phoenix tracing for live request debugging

Agent Memory Conflict-Resolution Eval Harness | *Python, BM25, FastAPI, Next.js, MongoDB, Redis, Docker*

- Engineered an ablation framework to evaluate a SOTA agent-memory conflict-resolution recipe (BM25 + LLM extraction) and isolate its true decision mechanism
- Designed a polarity-swap control experiment that identified relation-matching, not semantic reasoning, as the core mechanism driving conflict resolution
- Diagnosed the root cause behind multi-hop reasoning limitations and delivered a full-stack Memory Inspector demo (FastAPI, Next.js, MongoDB, Redis) for live evaluation

TECHNICAL SKILLS

- **Languages**: C++, C, Python, SQL
- **ML / Deep Learning**: Scikit-learn, PyTorch, TensorFlow, CNNs, LSTMs, Transformers, NLP
- **Generative AI**: LLM Fine-Tuning (LoRA, QLoRA), RAG, LangChain, Prompt Engineering, Diffusion Models
- **Libraries & Tools**: NumPy, Pandas, Matplotlib, FastAPI, Git, GitHub, Docker, Linux
- **CS Fundamentals**: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Computer Organization & Architecture

RELEVANT COURSEWORK

Foundations of Machine Learning | Deep Learning | Visual Computing | Linear Optimization | Fraud Analytics | Advanced Data Structures and Algorithms | Advanced Computer Networks | Network Security

POSITIONS OF RESPONSIBILITY

Teaching Assistant | Dept. of CSE, IIT Hyderabad

- Courses: Database Management Systems (CS3550), Operating Systems (CS3510), Introduction to AI (ID1050), Data Mining (CS5600)
- Assisted in coursework evaluation, resolved student queries, and helped maintain consistent grading standards